

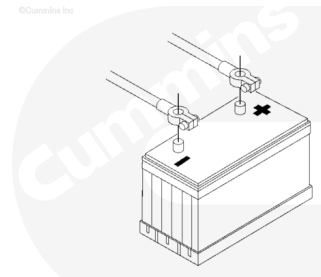
Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

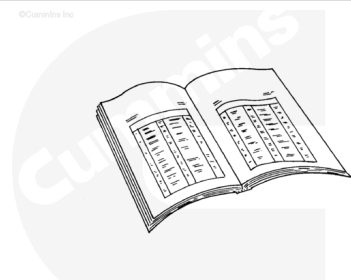
- Disconnect the batteries. See equipment manufacturer service information.



13900050

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.



ck800wa

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

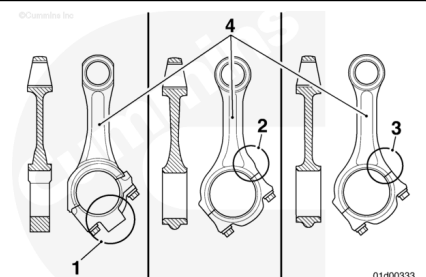
- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/40/40-007-037.html>)
- Remove the lubricating oil pan and gasket. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/40/40-007-025-tr.html>)
- If equipped, remove the block stiffener plate. Refer to Procedure 001-089 in Section 1. (</qs3/pubsys2/xml/en/procedures/40/40-001-089-tr.html>)
- If the engine is equipped with J-jet piston cooling nozzles, they **must** be removed to prevent component damage. Refer to Procedure 001-046 in Section 1. (</qs3/pubsys2/xml/en/procedures/40/40-001-046-tr.html>)
- Drain the coolant. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/40/40-008-018-tr.html>)
- Remove the cylinder head. Refer to Procedure 002-004 in Section 2. (</qs3/pubsys2/xml/en/procedures/40/40-002-004-tr.html>)

General Information

Different types of connecting rods are used in B Series engines, depending on the plant and time the engine was built.

Not all connecting rods are interchangeable. When replacing a connecting rod, make sure it matches the other connecting rods. All connecting rods in the engine **must** be the same.

Not all connecting rods have the part number located on the connecting rod. It is necessary to identify physical characteristics of the connecting rod when matching it to existing connecting rods.

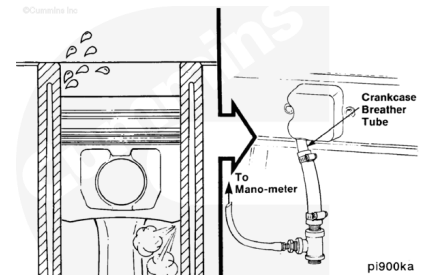


1. Balance weight on the connecting rod cap
2. Protrusion on short side of connecting rod
3. Smooth edge on the short side of the connecting rod

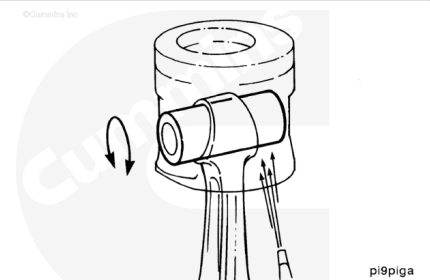
4. I-Beam design.

This procedure covers both the fracture-split (non-machined) and machined connecting rods. Cautions and notes are used throughout the procedure where special handling considerations **must** be followed for fracture-split (non-machined) rods.

There are a number of power-related problems, including excessive lubricating oil consumption, smoke, blowby, and poor performance, that can be caused by inadequate sealing between the piston rings and the cylinder walls. A blowby measurement can help detect the problem.



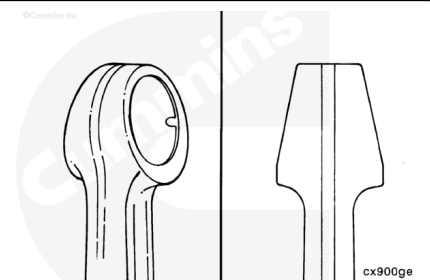
A free-floating, hollow piston pin is used to attach the piston to the connecting rod. Lubricating the pin and journal is accomplished by residual spray from piston cooling.



The piston pin end of the connecting rod is angle-cut to provide additional bearing surface. The connecting rod end is fitted with a bronze bushing.

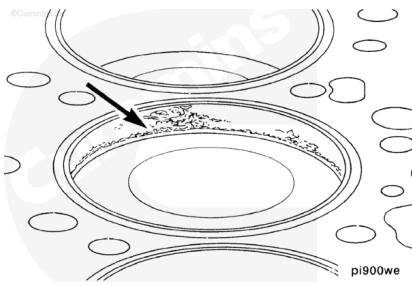
Some engines have connecting rods which contain a hole in the piston pin end of the connecting rod and the bushing. The hole in the rod and bushing could possibly **not** line up. The hole is for manufacturing purposes **only**.

Some four-cylinder naturally aspirated engines utilize a narrow lower connecting rod bearing.



Remove

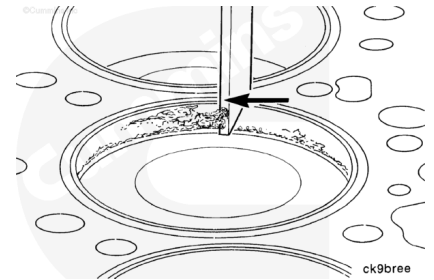
Rotate the crankshaft with engine barring tool, Part Number 3824591, until the pistons are below the carbon deposits, which are found above the ring travel area.



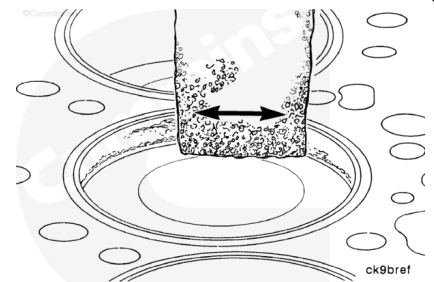
⚠ CAUTION ⚠

Do not use emery cloth or sandpaper to remove carbon from the cylinder bores. Aluminum oxide or silicon particles from these materials can cause serious engine damage.

Use a scraper or a blunt-edged instrument to loosen the carbon deposits. Do **not** damage the cylinder with the scraper.

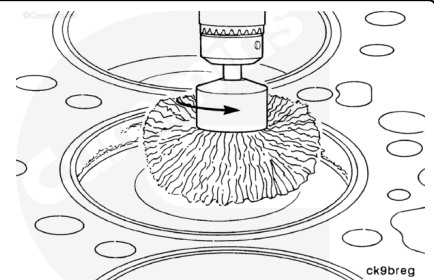


Remove the remaining carbon with a Scotch-Brite™ cleaning pad, or equivalent.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury



⚠ CAUTION ⚠

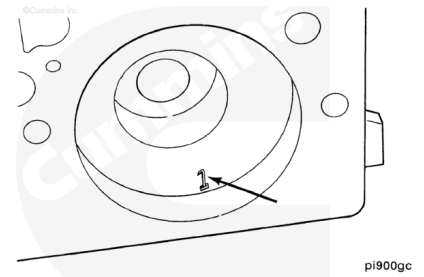
Do not use the steel wire wheel in the piston travel area. Operate the wheel in a circular motion to remove the deposits.

Note : An inferior quality wire wheel will lose steel bristles during operation, thus causing additional contamination.

An alternative method to remove the carbon ridge is to use a high-quality steel wire wheel installed in a drill or die grinder.

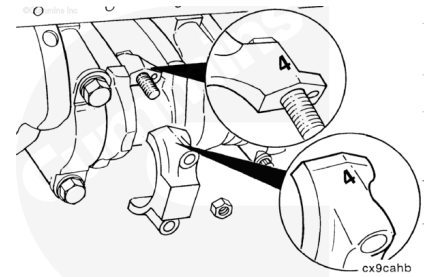
On pistons with anodized coatings, do **not** stamp the anodized coating on the outer rim.

Mark each piston according to the cylinder.



Rotate the crankshaft to position the connecting rod caps at bottom dead center for removal.

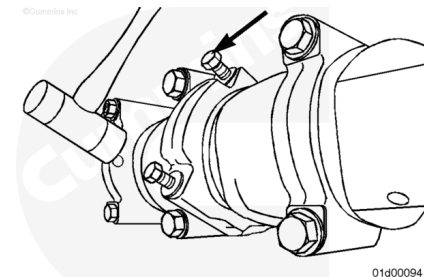
Mark each connecting rod and connecting rod cap according to the cylinder number location.



Loosen the connecting rod capscrews.

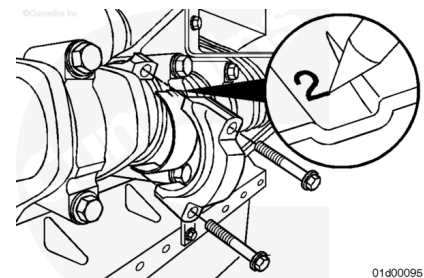
Note : Do not remove the capscrews from the connecting rods at this time.

Use a rubber hammer to hit the connecting rod capscrews to loosen the caps.



⚠ CAUTION ⚠

Do not damage the fracture-split surface on the connecting rod or connecting rod cap while the connecting rod caps are removed. If the fracture-split surface is damaged, the connecting rod and connecting rod cap must be replaced to help reduce the possibility of engine damage. Incorrect assembly can damage the rod.



⚠ CAUTION ⚠

When setting the connecting rod cap down do not set the cap down on the fracture-split surface. Damage to the fracture-split surface can result.

Remove the connecting rod capscrews.

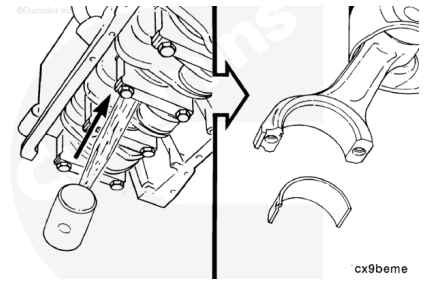
Remove the connecting rod cap.

Remove the lower connecting rod bearing.

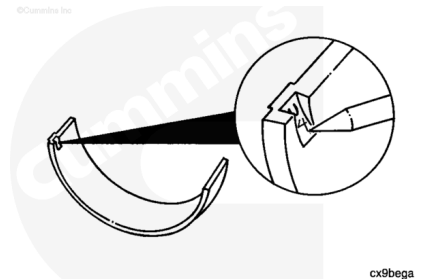
Mark the cylinder number and the letter "L" (lower) on the flat surface of the bearing tang.

Push the connecting rod and piston assembly out of the cylinder bore. Care **must** be taken **not** to damage the connecting rod or bearing.

Remove the upper rod bearing.



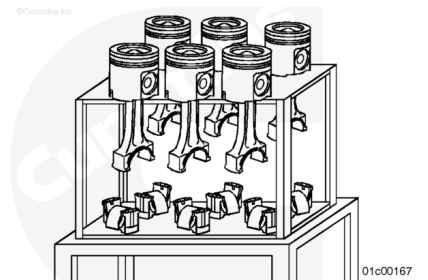
Mark the cylinder number and the letter "U" (upper) on the flat surface of the bearing tang.



The piston and connecting rod assemblies **must** be installed in the same cylinder number from which they were removed, to provide proper fit of worn mating surfaces, if parts are reused.

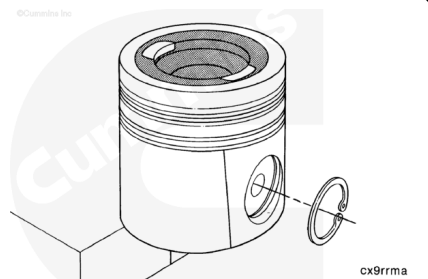
Use a tag to mark the cylinder number from which each piston and rod assembly were removed.

Place the rod and piston assemblies in a container to protect them from damage.



Disassemble

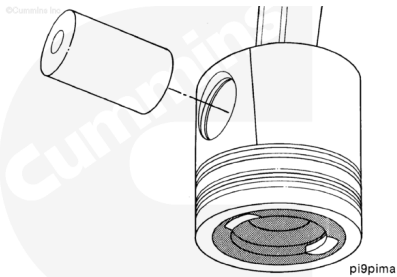
Remove the piston pin retaining rings.



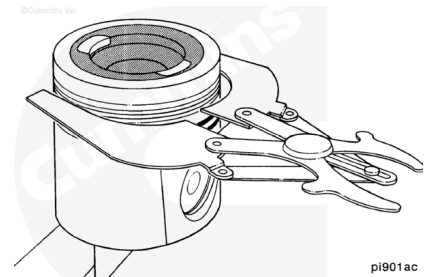
Note : Heating the piston is **not** required.

Remove the piston pin.

Remove the connecting rod from the piston.

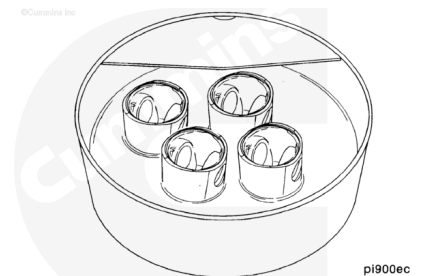


Remove the piston rings using piston ring expander, Part Number 3823137.

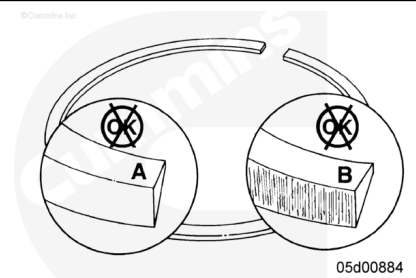


Clean and Inspect for Reuse

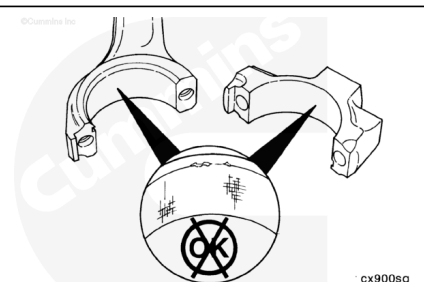
Clean and inspect the pistons and piston pins. Refer to Procedure 001-043 in Section 1. (</qs3/pubsys2/xml/en/procedures/40/40-001-043.html>)



Inspect the piston rings. Refer to Procedure 001-047 in Section 1. (</qs3/pubsys2/xml/en/procedures/40/40-001-047.html>)

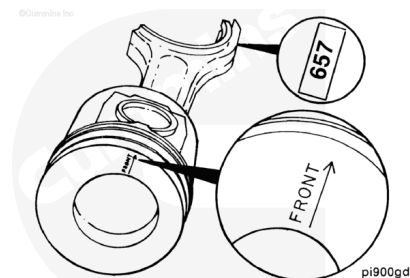


Inspect the connecting rod. Refer to Procedure 001-014 in Section 1. (</qs3/pubsys2/xml/en/procedures/40/40-001-014.html>)

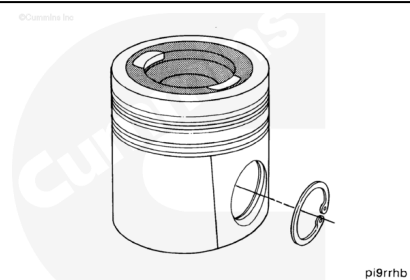


Assemble

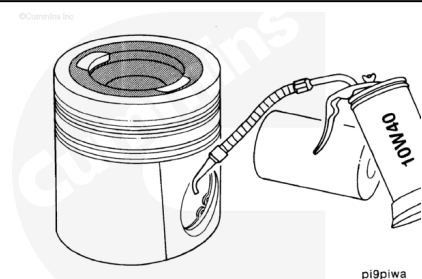
Be sure FRONT and/or arrow marking on the top of the piston and the numbers on the connecting rod and cap are positioned as illustrated.



Install the retaining ring in the pin groove on the front side of the piston.



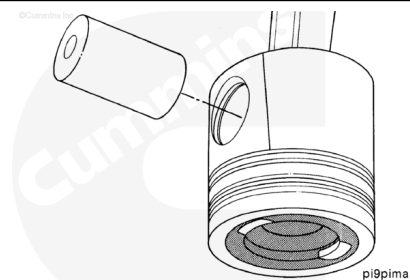
Lubricate the pin and pin bores with clean 15W-40 engine lubricating oil.



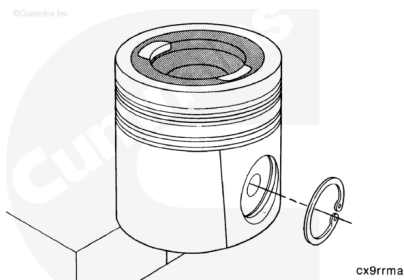
Note : Pistons do not require heating to install the pin; however, the pistons do need to be at room temperature or above.

Install the connecting rod.

Install the piston pin.



Install the second retaining ring.

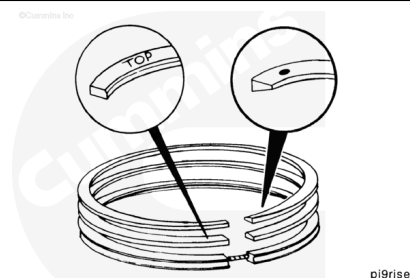


⚠ CAUTION ⚠

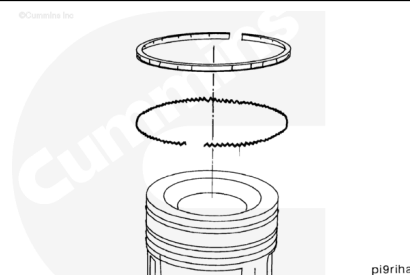
Most piston rings look similar but have significant differences. Make sure the correct part number is being used for the engine.

The top surface of the upper and intermediate rings are identified either with the word "TOP" or a supplier identification mark, such as a stamped dot. Assemble with the word "TOP" or the supplier mark facing upward.

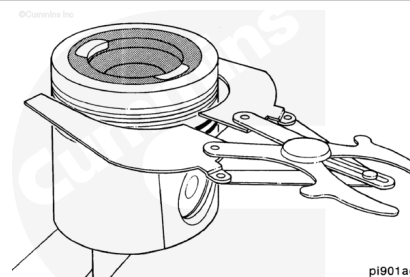
The bottom, or oil control ring, can be installed with either side up.



Note : The two-piece oil control ring must be installed with the expander ring gap 180 degrees from the gap of the oil ring.



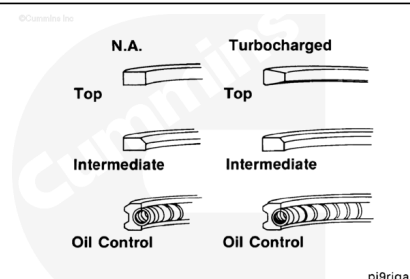
Using piston ring expander, Part Number 3823137, install the rings on the piston.



B3.9, B4.5, and B5.9 Engines

The piston ring type and location can be identified by piston ring profile.

1. Top piston ring
2. Intermediate piston ring

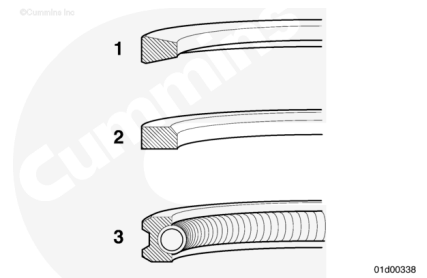


3. Oil control ring.

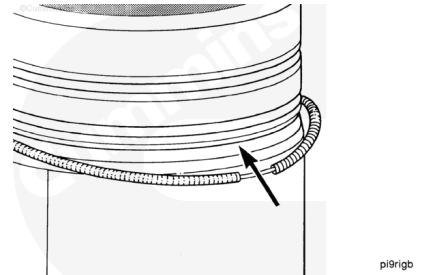
B4.5 RGT Engines

The piston ring type and location can be identified by piston ring profile.

1. Top piston ring
2. Intermediate piston ring
3. Oil control ring.



Start by positioning the oil ring expander in the oil control ring groove. Then install the oil control ring, followed by the intermediate ring, and finally the top ring.



Piston Grading

Note : Graded pistons were introduced into production in 1994. B4.5 RGT engines do not require piston grading.

When rebuilding an engine with the original cylinder block, crankshaft, and pistons, make sure the pistons are installed in the original cylinder.

If replacing the piston(s), make sure the replacement piston(s) are the same grade as the original piston.

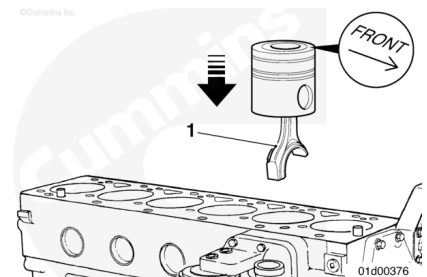
If a new cylinder block or crankshaft is used, the piston grading procedure **must** be performed to determine the proper piston grade for each cylinder.

Install the connecting rod and piston assembly into the number 1 cylinder without the rings installed.

Align the "Front" marking and/or arrow on the top of the piston so that it points toward the front of the engine.

Insert the connecting rod through the cylinder bore.

The long end of the connecting rod (1) will be on the exhaust side of the engine.



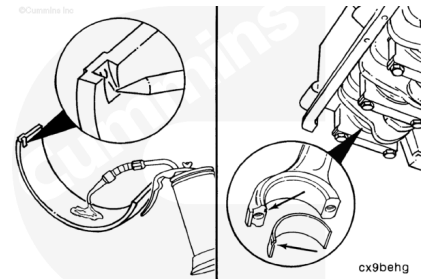
Note : The fracture surface on the fracture-split connecting rod and connecting rod cap must be kept

dry and clean to make sure proper mating of the two surfaces is maintained.

The connecting rod bearings **must** be installed in the original connecting rod and cap.

Install the upper shell in the connecting rod with the tang of the bearing in the slot of the connecting rod.

Use clean 15W-40 engine oil to coat the inside diameter of the connecting rod bearing shell.

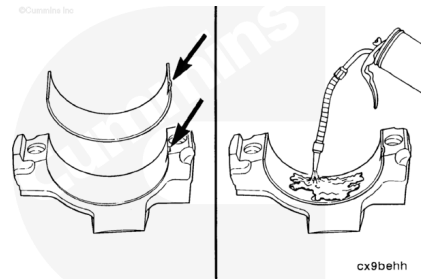


Install the bearing shell in the connecting rod cap with the tang of the bearing in the slot of the cap.

Note : The fracture-split connecting rods use a different upper and lower bearing.

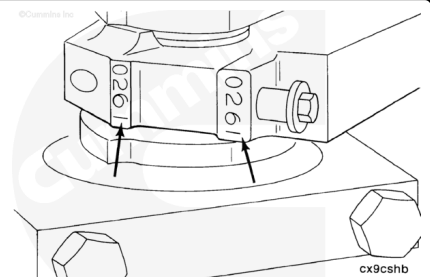
Use clean 15W-40 engine oil to coat the inside diameter of the connecting rod bearing shell.

Note : The fracture surface on the fracture-split connecting rod and connecting rod cap **must** be kept dry and clean to make sure proper mating of the two surfaces is maintained.



The number stamped on the connecting rod and cap at the parting line **must** match and be installed on the oil cooler side of the engine.

Install the connecting rod cap and capscrews to the connecting rod.

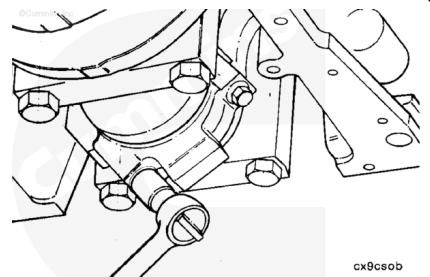


Tighten the two capscrews.

Torque Value:

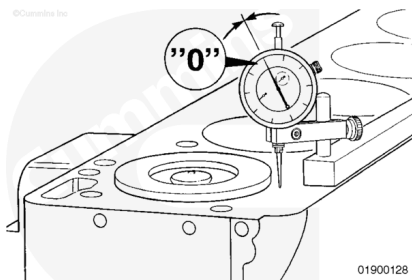
Connecting rod cap capscrews
1.35 n•m [26 ft-lb]

Do **not** tighten to final torque value.



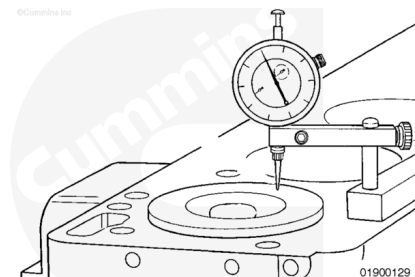
Use a fine-grit hone to remove any burrs from the cylinder block head deck.

Zero the depth gauge, Part Number 3164438, to the cylinder block head deck.



Note : Do not place the indicator tip on the anodized area.

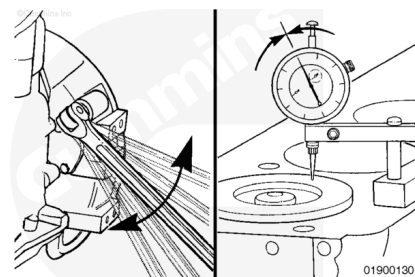
Move the depth gauge, Part Number 3164438, directly over the piston pin to eliminate any side-to-side movement.



Rotate the crankshaft to top dead center (TDC).

Rotate the crankshaft **clockwise** and **counterclockwise** to find the highest dial indicator reading.

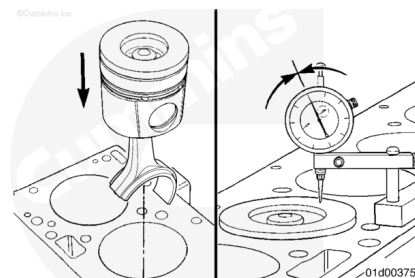
Record the reading.



Remove the piston and connecting rod assembly from the number 1 cylinder.

Install the assembly into the number 2 cylinder.

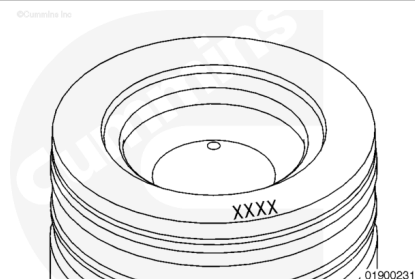
Repeat the procedure for every cylinder using the same piston and connecting rod assembly.



The four digits on top of the piston are the last four digits of the part number.

Using the measured piston protrusion and the grade of the piston, which was measured, determine the piston grade required to obtain protrusion from the following table.

The part number for each piston grade can be found by entering the engine serial number on QuickServe® Online Internet Website.



Piston Protrusion Chart			
Test Piston Grade	Protrusion		Use Grade
	mm	in	
A	0.609 to 0.711	0.024 to 0.028	A
A	0.508 to 0.609	0.020 to 0.024	B
A	0.406 to 0.508	0.016 to 0.020	C
B	0.711 to 0.813	0.028 to 0.032	A
B	0.609 to 0.711	0.024 to 0.028	B
B	0.508 to 0.609	0.020 to 0.024	C
C	0.813 to 0.914	0.032 to 0.036	A
C	0.711 to 0.813	0.028 to 0.032	B
C	0.609 to 0.711	0.024 to 0.028	C

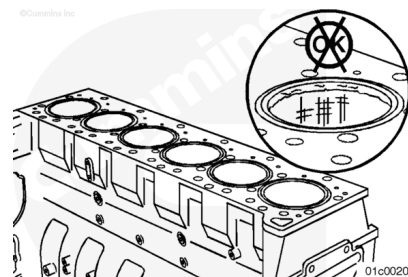
Install

The cylinder block and all parts **must** be clean before assembly.

Inspect the cylinder walls of the cylinder block. Refer to Procedure 001-026 in Section 1.

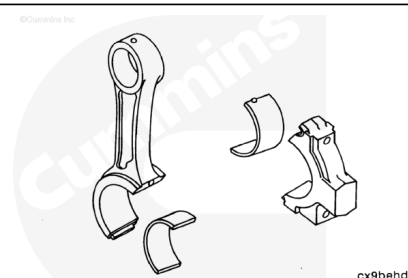
(/qs3/pubsys2/xml/en/procedures/40/40-001-026.html)

Use a clean, lint-free cloth to clean the connecting rods and bearing shells.



⚠ CAUTION ⚠

The connecting rods and connecting rod caps are not interchangeable. The connecting rods and connecting rod caps are machined as an assembly. Failure will result if the connecting rods and caps are mixed.



⚠ CAUTION ⚠

Make sure not to damage the mating surfaces of the fracture-split connecting rods and the rod caps.

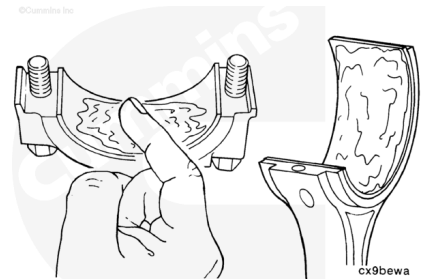
Note : Fracture-split connecting rods use a different upper and lower bearing.

Install the bearing shells into both the connecting rod and the connecting rod cap.

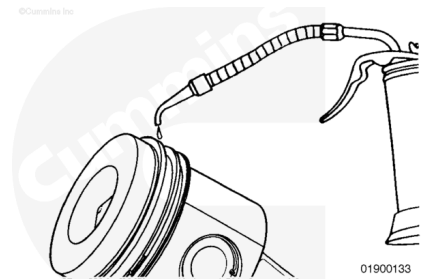
Make sure the tang on the bearing shells is in the slot of the connecting rod cap and connecting rod.

Note : The fracture surface on the fracture-split connecting rod and connecting rod cap must be kept dry and clean to make sure proper mating of the two surfaces is maintained.

Lubricate the connecting rod bearings with a light film of assembly lubricant, Part Number 3163087.

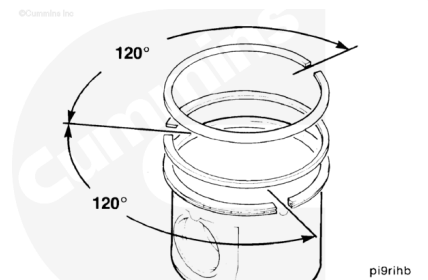


Lubricate the rings and piston skirts with clean engine lubricating oil.



Position the rings so that the ring gaps are 120 degrees apart.

Note : The ring gap of each ring must not be aligned with the piston pin, or any other ring. If the ring gaps are not aligned correctly, the rings will not seal properly.

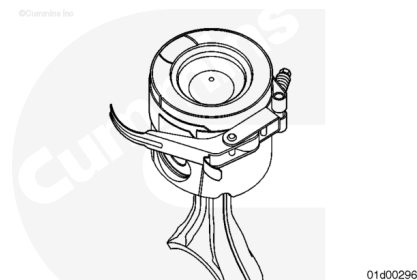


⚠ CAUTION ⚠

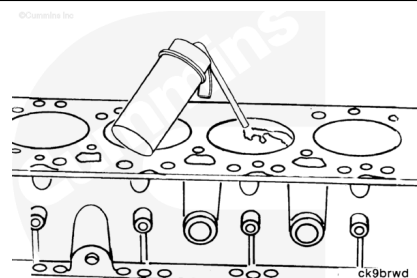
If using a strap type ring compressor, make sure the inside end of the strap does not hook on a ring gap and break the ring.

For B3.9 and B5.9 engines, use piston ring compressor, Part Number 3164330.

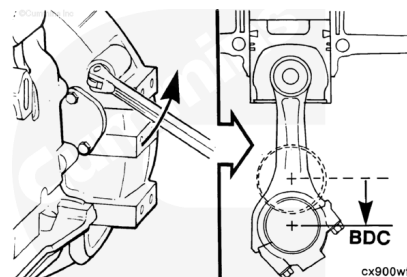
For B4.5 and B4.5 RGT engines, use piston ring compressor, Part Number 4918294.



Lubricate the cylinder bore with clean 15W-40 lubricating engine oil.



Position the connecting rod journal for the piston to be installed to bottom dead center (BDC).



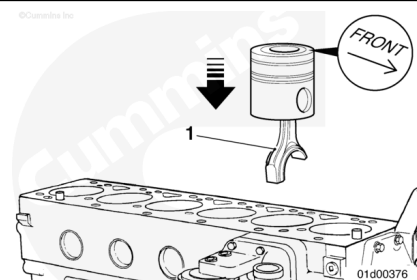
⚠ WARNING ⚠

Failure to follow this step will result in extensive engine damage.

Align the “front” marking and/or arrow on the top of the piston so that it points towards the front of the engine.

Insert the connecting rod through the cylinder bore until the ring compressor contacts the top of the cylinder block.

The long end of the connecting rod (1) will be on the exhaust side of the engine. If **not**, verify the piston is installed correctly onto the connecting rod.

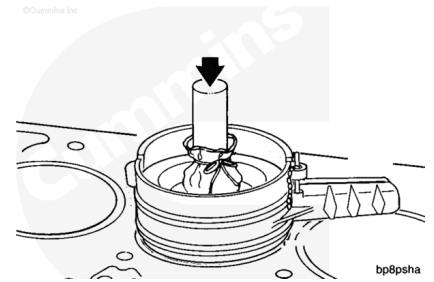


Hold the ring compressor against the cylinder block.

Push the piston through the ring compressor and into the cylinder bore.

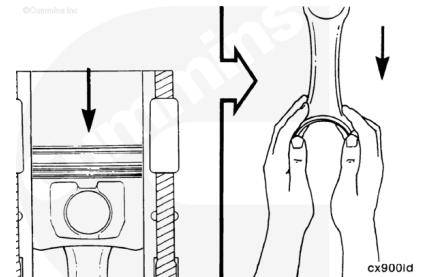
Push the piston until the top ring is completely in the cylinder bore.

Note : If the piston does not move freely, remove the piston and inspect for broken or damaged rings.



Take care to **not** damage the cylinder wall when inserting the connecting rod.

Carefully push the piston into the bore while guiding the connecting rod to the crankshaft journal.

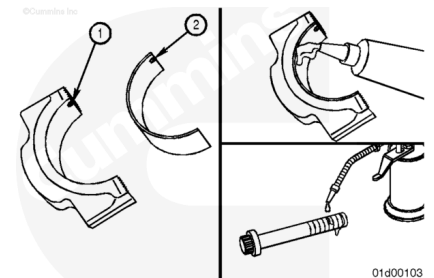


Note : The fracture surface on the fracture-split connecting rod and connecting rod cap **must** be kept dry and clean to make sure proper mating of the two surfaces is maintained.

Verify the tang of the bearing (2) is in the slot of the cap (1).

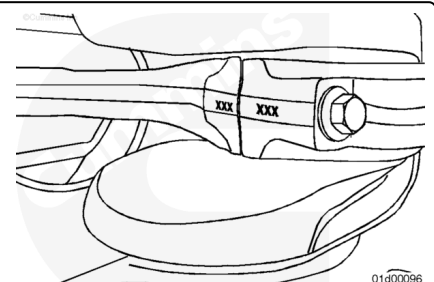
Use assembly lube, Part Number 3163087, or equivalent, to coat the inside diameter of the bearing shell.

Use clean 15W-40 oil to lubricate the connecting rod capscrew threads and underside of the connecting rod capscrew heads.



⚠ CAUTION ⚠

Do not damage the fracture-split surface on the connecting rod or connecting rod cap when the connecting rod cap is installed. If the fracture-split surface is damaged, the connecting rod and connecting rod cap must be replaced to help reduce the possibility of engine damage.



The connecting rod and cap **must** have the same number and **must** be installed in the proper cylinder. The connecting rod cap number and rod number **must** be on the same side of the connecting rod to prevent engine damage during engine operation.

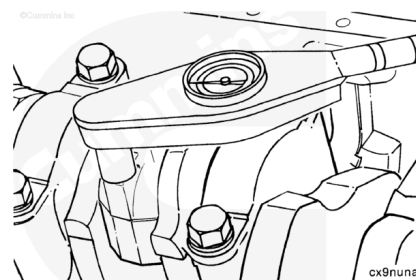
Install the connecting rod and capscrews.

Use a marked socket and torque wrench to tighten the connecting rod capscrews.

Torque Value:

B3.9, B4.5 RGT and B5.9 Engines

1. 45 $\text{n}\cdot\text{m}$ [33 $\text{ft}\cdot\text{lb}$]
2. Turn 90 degrees clockwise.



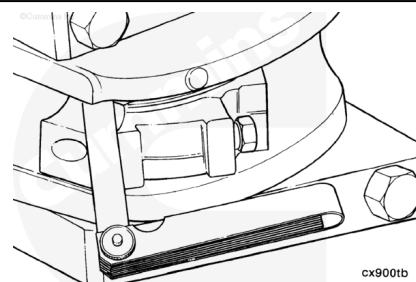
Torque Value:

B4.5 Engines

1. 25 $\text{n}\cdot\text{m}$ [221 $\text{in}\cdot\text{lb}$]
2. Turn 60 degrees clockwise.

Note : Do not measure the clearance between the connecting rod cap and crankshaft.

Measure the side clearance between the connecting rod and crankshaft.



Side Clearance Limits

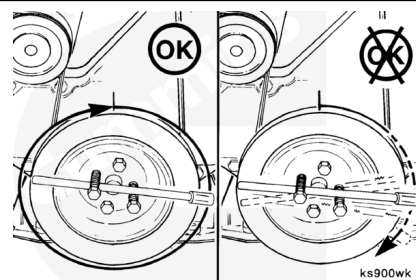
mm		in
0.10	MIN	0.004
0.33	MAX	0.013

⚠ CAUTION ⚠

To reduce the possibility of engine damage, the crankshaft must rotate freely.

⚠ CAUTION ⚠

If the connecting rod is not properly oriented (tang opposite the camshaft), it will contact the camshaft and lock the engine.



Check for freedom of rotation as the connecting rod caps are installed. If the crankshaft does **not** rotate freely, check the installation of the connecting rod bearings and the bearing size.

Measure

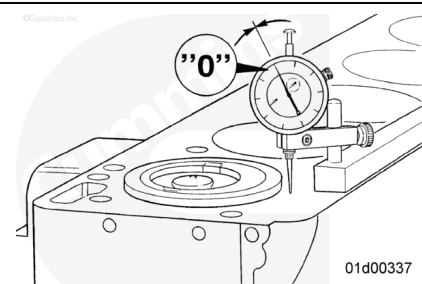
Measure piston protrusion above the cylinder block combustion deck.

Note : For B3.9, B4.5, and B5.9 engines, if piston grading procedure was performed, this measure step is **not** required.

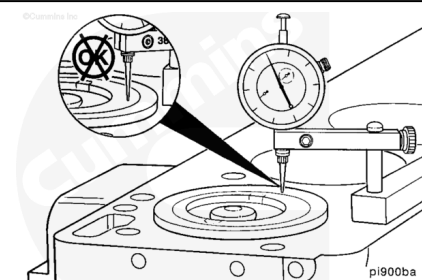
Note : For B4.5 RGT engines, this procedure is only required after a piston, crankshaft, connecting rod, or block modification/replacement.

Measure the piston protrusion using depth gauge assembly, Part Number 3823495. No piston or head gasket grading is required.

Install the dial indicator on the cylinder head and zero.



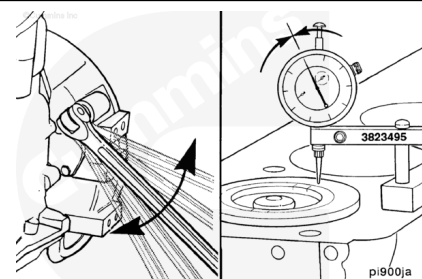
Move the dial indicator directly over the piston pin to eliminate any side-to-side movement. Do **not** place the indicator tip on the anodized area.



Rotate the crankshaft to top dead center. Rotate the crankshaft **clockwise** and **counterclockwise** to find the highest dial indicator reading. Record the reading.

B4.5 RGT Piston Protrusion		
mm		in
0.151	MIN	0.006
0.485	MAX	0.019

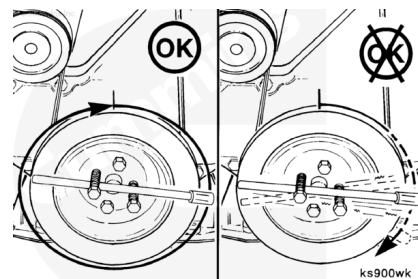
B3.9, B4.5, and B5.9 Piston Protrusion		
mm		in
0.609	MIN	0.024
0.711	MAX	0.028



If the piston protrusion is **not** within specification, verify that the correct parts are installed and/or the cylinder block combustion deck has been machined improperly. Refer to Procedure 001-026 in Section 1 (</qs3/pubsys2/xml/en/procedures/40/40-001-026.html>).

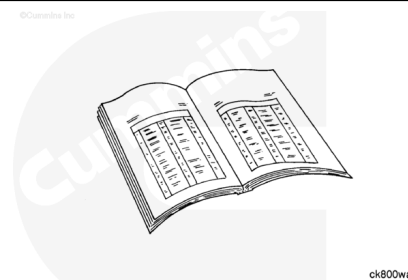
For B3.9, B4.5, and B5.9 engines verify the correct grade of piston is being used. See the Piston Grading section of this procedure.

Check for freedom of rotation as the connecting rod caps are installed. If the crankshaft does **not** rotate freely, check the installation of the connecting rod bearings and the bearing size.

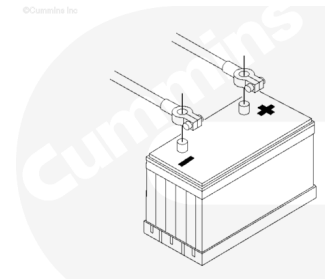


Finishing Steps

- If removed, install the J-jet piston cooling nozzles. Refer to Procedure 001-046 in Section 1. (</qs3/pubsys2/xml/en/procedures/40/40-001-046-tr.html>)
- Install the lubricating oil suction tube. Refer to Procedure 007-035 in Section 7. (</qs3/pubsys2/xml/en/procedures/40/40-007-035-tr.html>)
- If equipped, install the block stiffener plate. Refer to Procedure 001-089 in Section 1. (</qs3/pubsys2/xml/en/procedures/40/40-001-089-tr.html>)
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/40/40-007-025-tr.html>)
- Install the cylinder head. Refer to Procedure 002-004 in Section 2. (</qs3/pubsys2/xml/en/procedures/40/40-002-004-tr.html>)
- Fill the engine with lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/40/40-007-037.html>)
- Fill the engine with coolant. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/40/40-008-018-tr.html>).



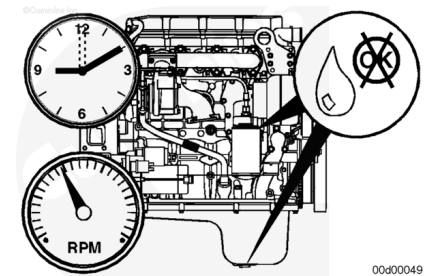
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



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- Connect the batteries. See equipment manufacturer service information.

- Operate the engine to normal operating temperature and check for leaks.



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